

Observation Patterns for Natural Language Organizational Analysis

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The Foundational Grammar: REA

Natural language observation of organizational life can be structured through three entity types drawn from accounting research and adapted for living systems. Together they provide a basic grammar for describing how organizations actually work.

Resources describe stocks of organizational capacity at a point in time, classified across seven forms of capital: financial, human, temporal, social, produced, natural, and cultural. A resource observation captures what exists, in what form, and at what level of vitality.

Events describe transformations that change resource stocks. Events can be classified as extractive (reducing capacity without replenishment), regenerative (enhancing future value creation), or reciprocal (balanced exchange). This classification reveals whether an organization is building resilience or depleting its foundation.

Agents are individuals, teams, or organizations with capacity for action and observation. Agency distributes through organizational structure rather than concentrating at the top; frontline workers observe dynamics invisible to senior leadership.

These three types function like subjects, verbs, and objects in a sentence. They allow description of what happened, who participated, and what changed. An observation enters the system as natural language, passes through semantic extraction to identify which entities are present, then enters human review before committing to organizational memory. This observe-extract-review-commit cycle is the metabolic rhythm underlying every pattern the system supports.

Primary Table Structure

Each entity type maps to a primary table with standard fields for multi-tenant isolation (OrgID), audit tracking (CreatedAt, UpdatedAt, CreatedBy), and referential integrity through UUID primary keys. Relations between tables use junction tables for many-to-many connections and foreign keys for direct associations.

Table	Purpose
Observations	Source table capturing raw natural language input, linked downstream to every entity extracted from it
Resources	Stocks with CapitalType, VitalityLevel, and temporal markers
Events	Transformations with EventClassification (extractive, regenerative, reciprocal), linked to participating Agents and affected Resources
Agents	People, teams, or organizations with AgentType and capability references

Computed columns surface derived insight without requiring additional input. Examples include resource vitality trends over time and event classification distributions per agent.

Extended Patterns and Their Tables

Building on the REA foundation, additional patterns can be deployed through a layered definition process. Each follows a five-layer derivation path: a Semantic Set defines meaning, a Data Schema defines structure, an Interface Design defines human interaction, a Semantic Processor defines extraction logic, and a Deployment Specification defines infrastructure.

Patterns that naturally emerge from organizational observation:

Pattern	Purpose
Profiles	Who participates and how they appear in organizational context. Extends the Agent table with capabilities, affiliations, and ecological relationships.
Projects	Coordinated activity with scope, timeline, and multi-capital resource allocation.
Offerings	What agents make available to others, with capacity tracking, availability windows, and matching infrastructure for coordination. Linked to source Observations for evidence.
Requests	The complement to Offerings; what agents need. Together, Offerings and Requests form a coordination layer.
Revisions	Enhancements to the system itself, documented through its own framework. This is where self-referential learning becomes operational.
Resolutions	Fixes and corrections, similarly self-documented.

Dialogue	A parallel entity type that preserves how understanding emerges through exchange, including texture markers, turning points, and vulnerability moments.
Gatherings	Documentation of when agents convene, capturing conditions and outcomes of collective activity.
Learnings	Capture of knowledge development through organizational experience.
Practices	Recurring activities that sustain organizational capability.
Stories	Narrative accounts that carry organizational meaning across time.

Operational Interface

The table structure enables three layers of operational interface.

Observation Entry accepts natural language input (voice, text, transcript upload) and routes it to the appropriate semantic processor. A routing layer sits between raw observation and pattern-specific processing, reading signals in natural language to determine which patterns apply. A single observation might trigger REA extraction, an Offering extraction, and a Profile update simultaneously. The router manages this multiplexing with confidence thresholds that balance automated processing against human judgment.

Review Workflow returns extracted entities to the observer for validation. Reviewers see source text alongside structured output, with options to accept, modify, reject, or defer. Reviewer corrections feed back into processor refinement, meaning the system learns what matters to each organization through accumulated feedback.

Accumulated Memory commits validated observations into queryable organizational knowledge. Cross-entity navigation surfaces patterns: which resources are changing, what events characterize organizational activity, and how agents participate across different contexts. Understanding builds through accumulation rather than isolated snapshots.

Progressive disclosure governs the interface throughout. Default views show what matters most. Expansion reveals additional detail. Expert modes remain available without cluttering everyday use. The goal is that reviewing an observation feels like reflection, not data entry.